Fundamental theorem of Morse Theory

- Let M be a Riemannian manifold with Riemann metric g, and let ρ be the induced topological metric. Let p and q be two points of M.

In homotopy theory, we study the pace space $\underline{\underline{\Omega^*}}$ of all continuous path

$$\omega : [0,1] \to M$$

from p to q, in the compact open topology. This topology can be also described as that induced by the metric

$$d^*(\omega, \omega') = \max_t \rho(\omega(t), \omega'(t))$$

$\underline{\underline{\Omega}}$ is the space of piecewise C^∞ paths from p to q with the metric

$$d(\omega, \omega') = d^*(\omega, \omega') + \left(\int_0^1 \left(\frac{ds}{dt} - \frac{ds'}{dt} \right)^2 dt \right)^{\frac{1}{2}}$$

For the natural map

$$i: \Omega \hookrightarrow \Omega^*$$

since $d \geq d^*$, i is continuous.

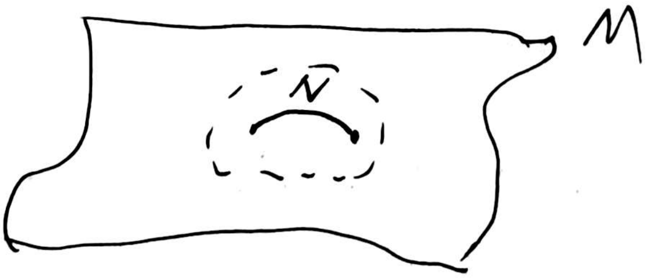
• Thm

This natural map i is a homotopy equivalence between Ω and Ω^*.

Proof:

By J.H.C Whitehead, every point of M has an open neighborhood N, which is "geodesically convex".

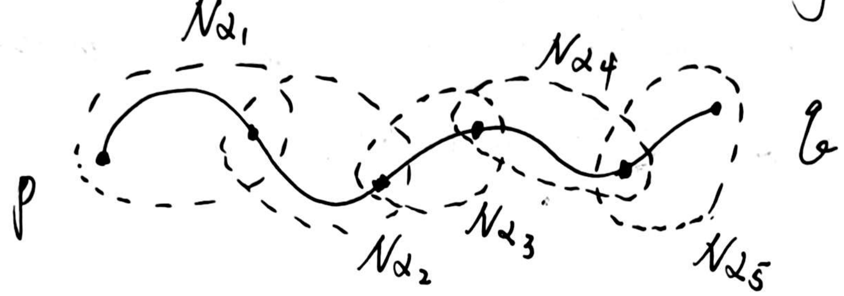
Any two points of N are joined by a unique minimal geodesic which lies completely within N and depends differentially on the endpoints.



(2)

Choose a covering of M by such geodesically convex open sets $\{N_\alpha\}$. Subdivid the interval $[0,1]$ into 2^k equal subintervals $[(j-1)/2^k, j/2^k]$, $j \in \{1, 2, \cdots, 2^k\}$. Let Ω_k^* denote the set of all continuous path w from p to q s.t.:

the image under w of each subinterval $[(j-1)/2^k, j/2^k]$ should be contained in one of the covering $\{N_k\}$.



So each Ω_k^* is an open subset of the space Ω^* of all paths from p to q, and Ω^* is the union of the sequence

$$\Omega_1^* \subset \Omega_2^* \subset \cdots$$

$$\Omega^* = \cup \Omega_k^*$$

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We define $\Omega_k = i^{-1}(\Omega_k^*)$. Similarly we have

$$\Omega_1 \subset \Omega_2 \subset \cdots$$

$$\Omega = \cup \Omega_k$$

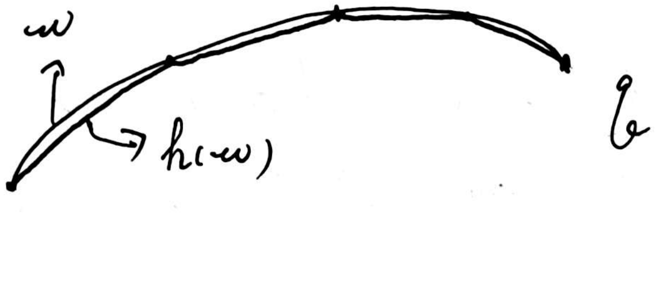
Since when k large enough, for $\forall w \in \Omega^*$, the image of w will be fine enough. Every segment will be covered by some N_α.

We first show that the natural map

$$(i|\Omega_k) : \Omega_k \to \Omega_k^*$$

is a homotopy equivalence.

For each $\omega \in \Omega_k^*$, let $h(\omega) \in \Omega_k$ be the broken geodesic which intersects with ω for the parameter values $t = j/2^k$, $j \in \{0, 1, \cdots, 2^k\}$, and which is a minimal geodesic within each intermediate interval $[(j-1)/2^k, j/2^k]$.



This define $h : \Omega_k^* \to \Omega_k$.

Lemma: $E' : B \to \mathbb{R}$ is smooth. (Morse theory, Milnor)

$B = \mathrm{Int}\,\Omega(t_0, t_1, \cdots, t_k)$ $\quad E_a^b(\omega) := \int_a^b \left(\frac{ds}{dt}\right)^2 dt$, $E := E_0^1$

$E' := E|_B$ $\quad \Omega^c := E^{-1}([0, c]) \subset \Omega$

$\mathrm{Int}\,\Omega^c := E^{-1}([0, c)) \subset \Omega$

$\Omega(t_0, t_1, \cdots, t_k) \subset \Omega$, s.t. $\omega|[t_{i-1}, t_i]$ is a geodesic for $\forall \omega$ and each i.

$\mathrm{Int}\,\Omega(t_0, t_1, \cdots, t_k)^c = \mathrm{Int}\,\Omega^c \cap \Omega(t_0, t_1, \cdots, t_k)$.

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Then, according to the lemma, $(i|_{\Omega_k})\circ h \simeq id_{\Omega_k^*}$ and $h\circ(i|_{\Omega_k}) \simeq id_{\Omega_k}$. So $i|_{\Omega_k}$ is a homotopy equivalence.

Lemma:

X is the homotopy direct limit of $\{X_i\}$ and Y is the homotopy direct limit of $\{Y_i\}$. Let $f: X \to Y$ be a map which carries each X_i into Y_i by a homotopy equivalence. Then f itself is a homotopy equivalence.

$$\begin{tikzcd} X_i \arrow[dr, "g_i"] \arrow[dd, "g_{ij}"'] & \\ & X \\ X_j \arrow[ur, "g_j"'] & \end{tikzcd} \qquad i \le j, \quad g_j \circ g_{ij} \simeq g_i$$

$$\begin{tikzcd} \Omega_1 \arrow[r, hook] \arrow[d, "i|_{\Omega_1}"] & \Omega_2 \arrow[r, hook] \arrow[d, "i|_{\Omega_2}"] & \cdots \arrow[r, hook] & \Omega_k \arrow[r, hook] \arrow[d, "i|_{\Omega_k}"] & \cdots \arrow[r, hook] & \Omega \arrow[d, "i"] \\ \Omega_1^* \arrow[r, hook] & \Omega_2^* \arrow[r, hook] & \cdots \arrow[r, hook] & \Omega_k^* \arrow[r, hook] & \cdots \arrow[r, hook] & \Omega^* \end{tikzcd} \qquad (5)$$

According to this lemma, $i: \Omega \hookrightarrow \Omega^*$ is a homotopy equivalence.

$\square$

- Thm (Milnor, On Spaces Having the Homotopy Type of a CW-Complex)

  If X is a compact CW-complex, $Y \in \mathcal{W}$, then, $Map(X, Y) \in \mathcal{W}$.

  $\mathcal{W}$: a space $X \in \mathcal{W}$, meaning $\exists$ a CW-complex K and a homotopy equivalence $f: K \to X$.

  Now consider Ω^* as a subset of $Map([0,1], X)$. As $Map([0,1], X) \in \mathcal{W}$, $\Omega^* \in \mathcal{W}$. So Ω^* has the homotopy type of a CW-complex.

$$\Omega \simeq \Omega^*$$

- Cor

  Ω has the homotopy type of a CW-complex.

- Thm (Fundamental theorem of Morse Theory)

  Let M be a complete Riemannian manifold, and let $p, q \in M$ be two points which are not conjugate along any geodesic. Then, $\Omega(M; p, q)$ has the homotopy type of a countable

CW-complex which contains one cell of dim λ for each geodesic from p to q of index λ.

Proof:

Choose a sequence $a_0 < a_1 < a_2 < \cdots$ of real number which are not critical values of the energy functional E, s.t. each interval (a_{i-1}, a_i) contains precisely on critical value. Consider the sequence

$$E(\omega) = \frac{1}{2}\int_0^1 g(\dot{\omega}(t), \dot{\omega}(t))\, dt$$

$$\Omega^{a_0} \subset \Omega^{a_1} \subset \Omega^{a_2} \subset \cdots$$

$$\Omega^{a_i} = \{\omega \in \Omega \mid E(\omega) \leq a_i\}$$

where Ω^{a_0} may be an empty set.

So each Ω^{a_i} has a homotopy type of $\Omega^{a_{i-1}}$ with a finite number of cells attached: one λ-cell for each geodesic of index λ in $E^{-1}(a_{i-1}, a_i)$.

Now we construct a sequence

$$K_0 \subset K_1 \subset K_2 \subset \cdots$$

of CW-complex.

Let B be the set of all broken geodesics from p to q. And $E' = E|_B$. According to our first lemma, $B^{a_0} = (E')^{-1}([0, a_0])$ is compact and it is a deformation retract of Ω^{a_0}. So $\Omega^{a_0} \simeq B^{a_0}$. We can choose K_0 to be a 0-dim CW-complex, s.t. $K_0 \simeq B^{a_0}$. So $K_0 \simeq \Omega^{a_0}$.

For $\forall i \geq 1$, we define K_i by induction. Let $c_i = E(\gamma) \in (a_{i-1}, a_i)$ be a critical value.

Every geodesic γ_j has Morse index λ_j. Every γ_j contributes a λ_j-cell, which is a λ_j-dim closed ball D^{λ_j} attached to K_{i-1} via $\varphi_j : S^{\lambda_i - 1} \to K_{i-1}$. So let

$$K_i = K_{i-1} \cup_{\varphi_j} \Big(\bigcup_{\gamma_j \text{ with } E(\gamma_j) \in (a_{i-1}, a_i)} D^{\lambda_j} \Big)$$

Then, $K = \bigcup K_i$. By induction, we can have $\Omega^{a_i} \simeq K_i$.

$f_i : \Omega^{a_i} \to K_i$ is the homotopy equivalence.

$f = \varinjlim f_i$

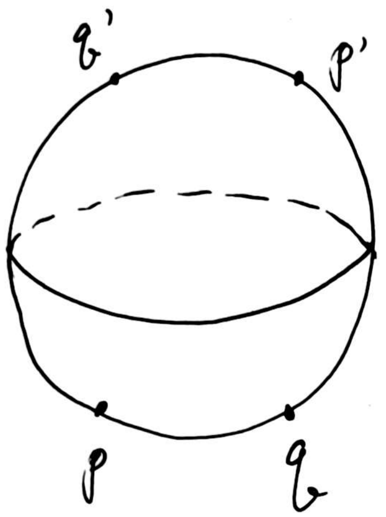
$$\begin{tikzcd}
\Omega^{a_0} \arrow[r, hook] \arrow[d, "f_0"', "\simeq"] & \Omega^{a_1} \arrow[r, hook] \arrow[d, "f_1"', "\simeq"] & \Omega^{a_2} \arrow[r, hook] \arrow[d, "f_2"', "\simeq"] & \cdots \arrow[r, hook] & \Omega^{a_i} \arrow[r, hook] \arrow[d, "f_i"', "\simeq"] & \cdots \Omega \arrow[d, "f"'] \\
K_0 \arrow[r, hook] & K_1 \arrow[r, hook] & K_2 \arrow[r, hook] & \cdots \arrow[r, hook] & K_i \arrow[r, hook] & \cdots K
\end{tikzcd}$$

So $f : \Omega \to K$ is a homotopy equivalence.

So Ω has the required homotopy type K. $\square$

• Example (The path space of S^n)

p, q are two non-conjugate points on S^n, $p \neq q$. p', q' are antipodes of p' and q', respectly. Let $\gamma_0, \gamma_1, \gamma_2, \cdots$ be geodesics from p to q.



Let $\gamma_0 = \overset{\frown}{pq}$

$\gamma_1 = \overset{\frown}{pq'p'q}$

$\gamma_2 = \overset{\frown}{pqp'q'pq}$

$\vdots$

The index $\lambda(\gamma_k) = \mu_1 + \cdots + \mu_k$ is equal to $k(n-1)$, as each of the points p or p' in the interior is conjugate to p with multiplicity $n-1$.

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• Cor

The loop space $\Omega(S^n)$ has the homotopy type of a CW-complex with one cell each in the dim $0, n-1, 2(n-1), 3(n-1), \cdots$

For $n>2$, the homotopy of $\Omega(S^n)$ can be computed immediately.

$$\begin{array}{ccccccccc} & & C_{k(n-1)-1} & & C_{k(n-1)} & & C_{k(n-1)+1} & & \\ \cdots & \to & 0 & \to & \mathbb{Z} & \to & 0 & \to & \cdots \end{array}$$